TwinVision Copper Pit

A digital open-pit copper mine on OpenEgiz

- 75 Eclipse Ditto twins: a live mine and a baseline copy
- Live 2D map, what-if scenarios with undo, decision modules with on/off switches
- Grade and hardness from the open GeoMet dataset, equipment from data sheets
- Mass, diesel and energy balances checked every step

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Contest "Digital Mine on OpenEgiz", 2026



The mine: from the face to copper in concentrate

3 block clusters → 3 shovels → 8 haul trucks on 12 road segments → primary crusher / ROM pad / waste dump → coarse ore stockpile → concentrator feed

Support systems

- **Dispatcher**: truck allocation, ore routing, blending, fuel rationing
- **Maintenance planner**: wear-dependent failure risk, planned stops
- **Fuel bay**: diesel stock, deliveries, refuelling queue
- **Power grid**: crusher and concentrator demand, 15-min peak, grid cap
- **Scenario controller**: what-ifs and module switches

Two copies

The **live** copy takes the what-ifs and switches. The **baseline** copy runs the same seed with every module on and no what-if. Each entity draws from its own random stream, so **live – baseline = the effect**.

```
mine                        Mine
  crusher-01                  Crusher
  dispatcher                  DispatcherController
  fuel-bay                    FuelBay
  maintenance                 MaintenancePlanner
  pit                         Pit
    blocks-east                 BlockCluster
    blocks-north                BlockCluster
    blocks-south                BlockCluster
    ex-01 … ex-03               Excavator ×3
    road-s01 … road-s12         RoadSegment ×12
    truck-01 … truck-08         HaulTruck ×8
  plant-feed                  PlantFeed
  power-grid                  PowerGrid
  scenario-controller         ScenarioController
  stockpile-cos               Stockpile
  stockpile-rom               Stockpile
  waste-dump                  WasteDump

+ the same tree under mine-baseline (37 twins)
  total 75 twins: all match SCHEMA.md
```

Output of `./twinvision.sh twins`: the hierarchy read from Ditto, checked against the inventory in SCHEMA.md.

Built on OpenEgiz: every value goes through the platform

Simulator (live + baseline, 1 s steps, 20× real time) → MQTT telemetry/<twin> → Eclipse Ditto (75 things) → MQTT opentwins/… → Telegraf → InfluxDB → Grafana

Map UI + control API ← Ditto REST (state, every 1 s) and InfluxDB (trends) · buttons and switches → scenario-controller twin in Ditto → simulator

- The UI never reads simulator memory: it polls Ditto for twin state and queries InfluxDB for history.
- What-if and module state live in a Ditto twin, so they survive restarts and anyone can set them with curl or a Ditto client.
- Twin geometry (benches, roads, blocks, facilities) is stored in twin attributes; the map draws it from there.
- One compose overlay on the platform. Start with `./twinvision.sh up`; `make` is not needed.

Footprint (measured on the running stack)

- Simulator about 20 MB, map backend about 41 MB of RAM
- Startup on a clean Ubuntu 24.04 host: 122 s including image pulls
- Map: plain HTML/SVG, no build chain, no GPU needed

Behaviour: physics, data and conservation

- **Trucks**: Cat 793F class (226.8 t, 1,848 kW). Speed comes from the power balance on each segment's grade and rolling resistance, with retarder limits downhill. Diesel is 0.25 L/kWh plus idle burn.
- **Shovels**: bucket size × fill × loose density, with swing cycles and spotting.
- **Ore**: block grades follow the 2,000 GeoMet copper assays and Bond work indices its 60 grindability tests. The spatial pattern is synthetic.
- **Energy**: Bond's law gives crusher kWh/t = 0.0129·Wi and concentrator kWh/t = 0.79·Wi, plus a 2.5 MW base load. Crusher capacity is limited by its 750 kW motor at the current hardness.
- **Recovery**: 87.9 %, the mean of the 53 GeoMet flotation tests.
- **Failures**: hazard λ0·exp(k(1 – health/100)), drawn from each machine's own random stream.
- **Queues**: shovels, crusher pocket, ROM and dump tip spots, fuel bay and workshop bays; roads have capacity limits.



The header shows the three balance residuals live, here during a crusher failure.

Every step: **mass** (dug + opening stocks – stocks, loads, waste, plant feed), **diesel** (opening + deliveries – burned, written off, tanks, bay) and **energy** (grid meter – fixed loads – Bond energy × tonnes). Both copies publish the residuals; the header shows them live.

46 unit tests: every residual below 0.000001 at every step, determinism, propagation of every scenario, each module on vs off.

Live mine map: http://localhost:8095/



The map refreshes every second from Ditto. Left: what-if scenarios and modules. Centre: the pit and plant, with trucks coloured by state and load-grade dots, blocks coloured by grade, and gauges with baseline markers. Right: live vs baseline KPIs, an InfluxDB trend and the event log.

What-if: crusher failure



- One click (or POST `/api/scenarios/crusher_failure`) writes the scenario-controller twin.
- The live crusher stops (red ring). The dispatcher sends ore to the ROM pad, the coarse ore stockpile drains, and plant feed and copper fall.
- The baseline keeps running; the KPI table shows the gap.
- The smoke test runs this round trip. In one run live crushing fell 584 t/h below the baseline.
- **Undo** restores the previous state, and the gap closes as the crusher catches up.
- **Resync live = baseline** copies the baseline state into the live copy, so the next what-if starts from identical mines.

Seven scenarios, all run from the UI or curl, all undoable



Road closure s04: the closed ramp is drawn red and trucks re-route.



Fuel shortage: the dispatcher parks trucks, waste first.

Scenario	Parameters	Effect vs baseline, as it appears on the stand
Crusher failure	–	ore to the ROM pad, coarse ore stockpile drains, plant feed and copper fall
Road closure	segment	re-routing; mined t/h down, diesel L/t up
Excavator breakdown	shovel	trucks re-assigned; queue time up, feed grade down
Hardness spike	factor, zone	slower digging and crushing; kWh/t up
Low-grade zone	factor, zone	feed grade and copper down; the dispatcher re-blends
Fuel shortage	resupply, stock	trucks rationed; mining drops, the plant runs on stockpiles
Energy cap	cap kW	concentrator throttled to the cap; plant feed down

Decision modules: logged decisions, on/off switch



- **Haulage and dispatch**: each empty truck goes to the shovel with the earliest expected loading start, corrected for the target-rate deficit and the feed grade. Ore goes to the ROM pad when the crusher is down or the grade is off band. Re-routing and fuel rationing are included. Off: fixed 2/2/4 allocation, all ore to the crusher.
- **Maintenance planner**: plans a stop when P(fail in 8 h) × MTTR × loss × 1.3 > PM time × loss, and builds coarse ore stock before crusher maintenance. Off: run to failure.
- **Energy manager**: under a grid cap it throttles the concentrator to stay below the cap. Off: a 15-min block over the cap trips the plant for 15 min.

Each decision is published to the module's twin (`decision_log`) with its reason. A switch writes the scenario-controller twin and affects the live copy only, so the KPI deltas show what the module is worth.



What each case changes over one simulated day

Output of `./twinvision.sh compare --hours 24`: the same simulator code run headless, seed 42, 2 h warm-up, then 24 h with the case applied to the live copy. Cells show live / baseline.

Case	Plant feed t	Grade % Cu	Copper t	Mined t	Queue min	Avail. %	kWh/t	Diesel L/t	Failures
crusher failure	3,613 / 47,666	0.715 / 0.634	22.7 / 265.4	174,888 / 173,136	0.47 / 0.85	87.6 / 95.0	32.05 / 17.47	0.204 / 0.215	2 / 2
road closure s04	47,369 / 47,666	0.623 / 0.634	259.3 / 265.4	161,870 / 173,136	0.65 / 0.85	94.4 / 95.0	17.30 / 17.47	0.235 / 0.215	2 / 2
excavator breakdown ex-01	33,542 / 47,666	0.542 / 0.634	159.8 / 265.4	145,363 / 173,136	4.99 / 0.85	87.5 / 95.0	17.62 / 17.47	0.200 / 0.215	1 / 2
hardness spike x1.4 north	44,146 / 47,666	0.637 / 0.634	247.1 / 265.4	171,736 / 173,136	0.80 / 0.85	95.1 / 95.0	21.41 / 17.47	0.211 / 0.215	2 / 2
low-grade zone x0.6 north	34,473 / 47,666	0.537 / 0.634	162.9 / 265.4	164,609 / 173,136	2.10 / 0.85	95.3 / 95.0	17.64 / 17.47	0.213 / 0.215	1 / 2
fuel shortage	32,977 / 47,666	0.647 / 0.634	187.5 / 265.4	38,944 / 173,136	0.07 / 0.85	97.6 / 95.0	17.82 / 17.47	0.313 / 0.215	0 / 2
grid cap 28 MW	37,590 / 47,666	0.649 / 0.634	214.3 / 265.4	172,462 / 173,136	0.79 / 0.85	94.7 / 95.0	17.88 / 17.47	0.209 / 0.215	2 / 2
dispatcher off	47,700 / 47,666	0.581 / 0.634	243.8 / 265.4	168,888 / 173,136	1.74 / 0.85	95.3 / 95.0	17.12 / 17.47	0.199 / 0.215	1 / 2
maintenance off	36,499 / 47,666	0.657 / 0.634	210.9 / 265.4	174,020 / 173,136	1.11 / 0.85	94.8 / 95.0	17.89 / 17.47	0.208 / 0.215	4 / 2
grid cap 28 MW, energy module off (48 load-shed events)	24,027 / 47,666	0.666 / 0.634	140.7 / 265.4	172,828 / 173,136	0.66 / 0.85	94.7 / 95.0	17.56 / 17.47	0.205 / 0.215	2 / 2

Module value: under the same 28 MW cap the energy manager keeps 214 t of copper against 141 t with load shedding. The dispatcher adds 22 t of copper per day and halves queue time. The maintenance planner prevents a crusher breakdown and saves 11,000 t of plant feed.

Grafana: live vs baseline from InfluxDB



Provisioned dashboard "TwinVision Copper Pit – live vs baseline" (http://localhost:3000/d/twinvision-overview). It pairs a live and a baseline series for every KPI and shows stockpiles, energy, diesel, balance residuals and maintenance.

twinvision/grafana/build_dashboards.py

Verified end to end

```
Containers
  ok    twinvision-setup created the twins
  ok    twinvision-sim healthy
Twins in Ditto
  ok    75 twins (live + baseline) indexed
  ok    twin mine, pit, crusher-01, ... exist (14 key twins)
  ok    truck-01 is a child of the pit (_parents)
  ok    twin types, counts, parents and relations match SCHEMA.md
Simulation
  ok    live mine twin advances (sim_time_h)
  ok    baseline mine twin advances
  ok    mine mass_balance_residual_t = 0.0 t
  ok    mine fuel_balance_residual_l = 0.0 L
  ok    mine energy_balance_residual_kwh = 0.0 kWh
  ok    mine-baseline mass_balance_residual_t = 0.0 t
  ok    mine-baseline fuel_balance_residual_l = 0.0 L
  ok    mine-baseline energy_balance_residual_kwh = 0.0 kWh
  ok    trucks move (x changes)
Telemetry in InfluxDB
  ok    points for mine in the last 2 min
  ok    points for mine-baseline in the last 2 min
  ok    points for truck-03 in the last 2 min
  ok    points for crusher-01 in the last 2 min
Grafana
  ok    TwinVision dashboard provisioned
Web UI
  ok    twinvision-web healthy
  ok    map page served at http://127.0.0.1:8095
  ok    map geometry from twin attributes (/api/static)
  ok    live state from Ditto (/api/state)
  ok    trend from InfluxDB (/api/trend)
What-if round trip (crusher failure, then undo)
  ok    scenario started through the web API
  ok    live crusher failed
  ok    baseline crusher unaffected
  ok    live crusher throughput 0 t/h
  ok    live tph_crushed below baseline by > 500 t/h
        delta vs baseline after the failure: -584.2 t/h
  ok    undo through the web API
  ok    live crusher back in service
  ok    live crushing KPI recovers (tph_crushed)
  ok    gap to baseline closes (tph_crushed)
        delta vs baseline after undo: -531.0 t/h
Decision module switch (maintenance off, then undo)
  ok    maintenance module switched off through the web API
  ok    live maintenance planner reports module off
  ok    baseline maintenance planner stays on
  ok    live twin lists the module as off (mine modules_off)
  ok    undo through the web API
  ok    live maintenance planner back on
Resync live to baseline
  ok    resync requested through the web API
  ok    live and baseline report identical totals (mined_total_t)
TwinVision smoke test passed.
```

Run it

```
git clone https://github.com/openegiz-contest/team-038.git
cd team-038
./twinvision.sh up        # platform + TwinVision
./twinvision.sh smoke     # end-to-end check
./twinvision.sh test      # 46 unit tests in the image
```

- A clean Ubuntu 24.04 host (Docker 29.1, Compose 2.40, no `make`) needs no manual steps: 122 s to start, then all checks pass.
- The platform's own `make smoke` also passes.

Limitations

- 2D map only; there is no Unity scene and no vision module.
- The concentrator is a feed sink with Bond energy and constant recovery.
- The site is fictional. Only the grade, hardness and recovery distributions come from real data.
- The simulator restarts from t = 0 with its container; scenario state persists in Ditto.